Light

GCSE Computer Science - Cyber Security Threats (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which of the following is most likely to indicate a malware infection on a computer?

- ○ **A.** The computer suddenly runs much slower than usual and unknown programs appear
- ○ **B.** The computer requires a password to log in
- ○ **C.** Software updates are available for download
- ○ **D.** The screen brightness adjusts automatically

Q2 (1 mark)

Which password would be most vulnerable to attack?

- ○ **A.** T8#mK2@pL9!x
- ○ **B.** 123456
- ○ **C.** c0mP1exP@55w0rd!
- ○ **D.** MyD0g$Name2024!

Q3 (1 mark)

What is a 'payload' in the context of malware?

- ○ **A.** The amount the victim must pay in a ransomware attack
- ○ **B.** The weight of the infected computer
- ○ **C.** The number of files on the computer
- ○ **D.** The malicious action or code that the malware delivers and executes

Q4 (1 mark)

Why do social engineering attacks often create a sense of urgency?

- ○ **A.** Because attackers are in a hurry to finish
- ○ **B.** Because urgent messages are easier to write
- ○ **C.** Because urgency pressures victims to act quickly without thinking critically
- ○ **D.** Because computers process urgent requests faster

Q5 (1 mark)

Why are unpatched systems particularly vulnerable to cyber attacks?

- ○ **A.** Unpatched systems have known security flaws that attackers can exploit
- ○ **B.** Patches make computers run slower
- ○ **C.** Patching removes all security features
- ○ **D.** Unpatched systems have newer software

Q6 (1 mark)

What is credential harvesting?

- ○ **A.** Creating strong passwords for users
- ○ **B.** Growing vegetables using computer-controlled systems
- ○ **C.** Collecting usernames and passwords through deceptive methods
- ○ **D.** Deleting old user accounts from databases

Q7 (1 mark)

Why do ransomware attackers typically demand payment in cryptocurrency?

- ○ **A.** Cryptocurrency transactions are anonymous and difficult to trace back to the attacker
- ○ **B.** Cryptocurrency is the only form of digital money

- ○ **C.** Banks do not allow ransomware payments
- ○ **D.** Cryptocurrency payments are always refundable

Q8 (1 mark)

Which type of organisation is most commonly targeted by cyber attacks seeking financial data?

- ○ **A.** Libraries
- ○ **B.** Financial institutions and retailers
- ○ **C.** Weather forecasting services
- ○ **D.** Public parks

Q9 (1 mark)

What does 'propagation' mean in the context of malware?

- ○ **A.** How malware spreads from one system to another
- ○ **B.** How malware is deleted from systems
- ○ **C.** How malware improves computer performance
- ○ **D.** How malware displays error messages

Q10 (1 mark)

What is reconnaissance in the context of a cyber attack?

- ○ **A.** Gathering information about a target before launching an attack
- ○ **B.** Deleting evidence after an attack
- ○ **C.** Repairing systems after a breach
- ○ **D.** Training employees about security

Q11 (1 mark)

What is the main impact of a successful DoS attack on a business website?

- ○ **A.** Customer data is encrypted and held for ransom
- ○ **B.** The website becomes unavailable to legitimate users, causing loss of business
- ○ **C.** All employee passwords are changed
- ○ **D.** The website displays different content

Q12 (1 mark)

Which part of a website is typically exploited in an SQL injection attack?

- ○ **A.** User input fields such as login forms and search boxes
- ○ **B.** The website's colour scheme
- ○ **C.** The website's loading animation
- ○ **D.** The website's privacy policy page

Q13 (2 marks)

A computer starts displaying pop-up advertisements even when the browser is closed, and the user notices their homepage has changed without permission.
What type of malware behaviour does this indicate?

- ○ **A.** Ransomware - the pop-ups are ransom demands
- ○ **B.** Worm - it is spreading to other computers
- ○ **C.** Adware and browser hijacker - displaying unwanted ads and modifying browser settings without consent
- ○ **D.** Keylogger - recording the advertisements viewed

Q14 (2 marks)

An email claims to be from a delivery company about a missed parcel. It contains spelling errors, asks the user to click a link to reschedule, and the sender address is 'delivry-notice@randomsite.net'.
What evidence suggests this is a phishing attempt?

- ○ **A.** Delivery companies never send emails
- ○ **B.** The spelling errors, suspicious sender address, and request to click a link are classic phishing indicators
- ○ **C.** The email mentions a parcel
- ○ **D.** Phishing emails are always about banking

Q15 (2 marks)

In a DDoS amplification attack, attackers send small requests to servers that respond with much larger replies, all directed at the victim.
Why is this technique effective?

- ○ **A.** Larger replies take longer to delete
- ○ **B.** Servers prefer responding to small requests
- ○ **C.** It multiplies the attack traffic volume without requiring equivalent attacker bandwidth, overwhelming the target more efficiently
- ○ **D.** Amplification makes the requests invisible

Q16 (2 marks)

An attacker calls an employee pretending to be the company CEO, demanding an urgent bank transfer to a new supplier account.
What social engineering principle is being exploited?

- ○ **A.** Authority - people tend to comply with requests from perceived authority figures without questioning
- ○ **B.** Scarcity - there are not enough bank accounts
- ○ **C.** Reciprocity - the employee owes the CEO a favour
- ○ **D.** Consensus - everyone else has made the transfer

Q17 (2 marks)

A company suffers a data breach where customer names, email addresses, and encrypted passwords are stolen.
Why are affected customers still at risk even though passwords were encrypted?

- ○ **A.** Encryption makes passwords easier to read
- ○ **B.** Email addresses cannot be encrypted
- ○ **C.** Names are more valuable than passwords

○ **D.** Attackers can attempt to crack the encryption, and customers who reuse passwords on other sites may have those accounts compromised too

Q18 (2 marks)

After cleaning a malware infection, the malware reappears every time the computer restarts.
What characteristic of the malware enables this behaviour?

○ **A.** The malware is stored in RAM

○ **B.** The malware has established persistence mechanisms that allow it to survive reboots and reinstall itself

○ **C.** The computer's power supply is infected

○ **D.** Malware cannot actually be removed

Q19 (2 marks)

An attacker on the same unsecured Wi-Fi network uses packet sniffing software to capture data transmitted by other users.
What type of data is most vulnerable to this attack?

○ **A.** Data stored on the user's hard drive

○ **B.** Data that has already been deleted

○ **C.** Data stored in cloud backups

○ **D.** Unencrypted data transmitted over the network, such as HTTP traffic and plaintext emails

Q20 (2 marks)

A security audit finds that employees frequently write passwords on sticky notes attached to their monitors.
What type of vulnerability does this create?

- ○ **A.** A software vulnerability in the operating system
- ○ **B.** A network vulnerability that can be exploited remotely
- ○ **C.** A physical security vulnerability where anyone who can see the monitor can access the password
- ○ **D.** A hardware vulnerability in the monitor

Q21 (2 marks)

Attackers compromise a popular industry news website that their target organisation's employees frequently visit. Visitors to the site are silently infected with malware.
What type of attack is this?

- ○ **A.** Spear phishing - targeting specific employees with emails
- ○ **B.** Brute-force - trying all password combinations
- ○ **C.** Watering hole attack - compromising websites that targets are known to visit
- ○ **D.** Denial of service - making the website unavailable

Q22 (2 marks)

An attacker successfully performs SQL injection on a web application and executes the command: DROP TABLE customers;
What is the impact of this command?

- ○ **A.** It displays all customer information on screen
- ○ **B.** It creates a backup of customer data
- ○ **C.** It improves database performance
- ○ **D.** It permanently deletes the entire customers table and all data within it from the database

Q23 (2 marks)

Attackers compromise a software company's update server. When customers download the latest update, they unknowingly install malware alongside the legitimate software.
What type of attack is this?

- ○ **A.** Phishing attack - tricking users with fake emails
- ○ **B.** Brute-force attack - guessing server passwords
- ○ **C.** Supply chain attack - compromising trusted software distribution channels to spread malware
- ○ **D.** Social engineering - manipulating individual employees

Q24 (3 marks)

A sophisticated attack on a company proceeds through these stages: researching employees on LinkedIn, sending a targeted email with a malicious attachment, the malware establishing a backdoor, the attacker moving through the network to find sensitive data, and finally exfiltrating customer records.
Analyse the attack lifecycle stages demonstrated.

- ○ **A.** This is a single-stage attack using only social engineering

○ **B.** Reconnaissance (LinkedIn research), initial access (spear phishing), persistence (backdoor), lateral movement (network exploration), and exfiltration (data theft) - a complete attack chain

○ **C.** The attack failed because it used too many stages

○ **D.** Only the final data theft stage matters; earlier stages are irrelevant

Q25 (3 marks)

A security assessment identifies these issues: a web server running software from 2019 with known vulnerabilities, employees using their pet names as passwords, and a firewall with default administrator credentials.
Analyse the vulnerability categories represented and their relative risks.

○ **A.** These are all hardware problems requiring physical replacement

○ **B.** Only the firewall issue matters because firewalls are for security

○ **C.** Pet names are strong passwords because they are personal

○ **D.** Software vulnerability (outdated server), human vulnerability (weak passwords), and configuration vulnerability (default credentials) - all high risk as each provides potential attack entry points

Q26 (3 marks)

A manufacturing company's production systems are encrypted by ransomware. Production stops completely, customer orders cannot be fulfilled, the company cannot process payments, and employees cannot access work systems.
Analyse the business impacts of this attack.

○ **A.** The only impact is that files are encrypted

○ **B.** Operational impact (production stops), financial impact (lost revenue, unfulfilled orders), reputational impact (customer trust), and workforce impact (employees cannot work) - cascading effects across the entire business

○ **C.** Manufacturing companies cannot be affected by ransomware

○ **D.** Ransomware only affects personal computers, not business systems

Q27 (3 marks)

A company implements technical controls including email filtering and web blocking. However, employees still fall for social engineering attacks delivered through phone calls and in-person visits.
Analyse why technical controls alone are insufficient against social engineering.

○ **A.** Technical controls are completely useless against all attacks

○ **B.** Phone calls cannot be used for attacks

○ **C.** Technical controls should have blocked the phone calls

○ **D.** Social engineering exploits human psychology through multiple channels; technical controls only filter digital communications but cannot prevent voice calls, physical access, or psychological manipulation of authorised users

Q28 (3 marks)

Modern malware often uses techniques such as: encrypting its code, changing its signature with each infection, only activating when not in a sandbox environment, and mimicking legitimate system processes.
Analyse how these techniques help malware evade detection.

○ **A.** These techniques make malware easier to detect

○ **B.** Encryption hides malicious code from scanners, polymorphism defeats signature matching, sandbox detection avoids analysis, and process mimicry blends with legitimate activity - each technique counters a specific detection method

○ **C.** Antivirus software cannot be evaded by any technique

○ **D.** Only the encryption technique provides any evasion benefit

Q29 (3 marks)

An attacker wants to compromise a specific company. They consider three approaches: mass phishing emails to all employees, exploiting a known vulnerability in the company's public web server, or physically entering the building with a fake contractor badge.
Analyse the advantages and risks of each attack vector.

○ **A.** All three approaches are equally effective and carry identical risks

○ **B.** Physical access is impossible because buildings have locks

○ **C.** Only technical attacks like web exploitation can succeed

○ **D.** Mass phishing has high volume but low success rate and leaves evidence; web exploitation is technical but quick if vulnerable; physical access provides direct access but highest personal risk - each vector has distinct trade-offs

Q30 (3 marks)

A hospital network is compromised. Investigation reveals: initial access came through a phishing email to a receptionist, the malware spread to medical devices through the internal network, patient monitors began displaying false readings, and attackers demanded payment to restore normal operation.
Analyse the attack components and why healthcare is a high-value target.

- ○ **A.** Hospitals are never targeted by cyber attacks
- ○ **B.** Only the payment demand was relevant; other stages had no impact
- ○ **C.** Medical devices cannot be affected by cyber attacks
- ○ **D.** The attack combined phishing (initial access), worm-like propagation (network spread), sabotage (false readings creating patient safety risk), and ransomware (payment demand) - healthcare is high-value because patient safety creates urgency to pay

Total: **55** marks • Time: **30 min**

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